

Create Rubric

25 points

Create your rubric now or come back to it later. You can also make edits to your rubric while grading.

Q1 Question 1ab

2 points

Rubric !

v. A (page 2 of 6)
1. Let $f(x) = \frac{x}{x+2}$ and $g(x) = x - 5$. Find:
(a) (1 point) $f(g(6)) =$

Name: _____

(b) (1 point) $f(f(1)) =$

(c) (2 points) $f(g^{-1}(5)) =$

(d) (2 points) $f(g(x)) =$

1 +1.0
Correct 1a) $\frac{1}{3}$

2 +1.0
Correct 1b) $\frac{1}{7}$

3 +0.0
Incorrect

4 +2.0
Different Version

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Q2 Question 1cd

4 points

Rubric !

(c) (2 points) $f(g^{-1}(5)) =$

(d) (2 points) $f(g(x)) =$

1 -0.0
Correct

2 -1.0
 $g^{-1}(5)$ incorrect

3 -2.0
part (c) incorrect

4 -2.0
part (d) incorrect

5 -1.0
did not compute $f(10)$

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Q3 Question 2ab

2 points

Rubric!

v. A (page 8 of 6)

2. (a) (1 point) What is the domain of $\cos^{-1}(x)$?

(b) (1 point) What is the range of $\cos^{-1}(x)$?

(c) (4 points) Find $\sin\left(-\cos^{-1}\left(\frac{1}{2}\right)\right)$.

Name: _____

1

-0.0

Correct

2

-1.0

part a incorrect

3

-1.0

part b incorrect

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Q4 Question 2c

4 points

Rubric!

1

+1.0

$\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$

2

+1.0

$\cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$ is unique. That is $\cos^{-1}\left(\frac{1}{2}\right)$ has no values in its range

3

+1.0

no sign error

4

+1.0

$-\frac{\sqrt{3}}{2}$: or off by sign

5

+0.0

Both parts incorrect /No work

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Q5 Question 3a

4 points

 Rubric!

v. A (page 4 of 6)

Name: _____

3. (6 points) Let $f(x) = x^4 \sin\left(\frac{1}{x}\right)$. For each of the following limits, determine if they exist and if it does, determine its value. Be sure to show how you arrived at your answer.

(a) $\lim_{x \rightarrow 0} f(x)$

(b) $\lim_{x \rightarrow \frac{\pi}{2}} f(x)$

 1 -0.0

Completely correct:
 $\lim_{x \rightarrow 0} f(x) = 0$ using Squeeze Theorem

 2 -2.0

Did not include inequality
 $-1 \leq \sin(1/x) \leq 1$

 3 -2.0

Did not employ squeeze theorem

 4 -1.0

Other minor issue (see comments)

 5 -3.5

Incorrect approach

Common incorrect approaches included:

1. Plugging in $x = 0$ and determining that the limit not exist since $\sin(1/0)$ is undefined.
2. Plugging in $x = 0$ and incorrectly stating that $\sin(1/0) = 0$ and thus concluding that the limit is

 6 -4.0

Blank/No attempt or insufficient work shown to receive credit

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Q6 Question 3b

2 points

 Rubric!

v. A (page 4 of 6)

Name: _____

3. (6 points) Let $f(x) = x^4 \sin\left(\frac{1}{x}\right)$. For each of the following limits, determine if they exist and if it does, determine its value. Be sure to show how you arrived at your answer.

(a) $\lim_{x \rightarrow 0} f(x)$

(b) $\lim_{x \rightarrow \frac{\pi}{2}} f(x)$

 1 +2.0

Correct w/ work

 2 +1.0

Incorrect w/ work

 3 +0.0

Insignificant progress/missing

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Q7 Question 4

6 points

Rubric!

v. A (page 5 of 6)

Name: _____

4. (6 points) Consider the function $f(x) = \begin{cases} \frac{x^2-4}{x-2} & \text{if } x < 2 \\ ax & \text{if } x \geq 2 \end{cases}$

(a) What is the value of $f(2)$?

(b) Determine the value of a for which $f(x)$ is continuous for all x . Be sure to show the limit you evaluated to arrive at your answer.

1 -0.0

Correct

2 -2.0

(a)

3 -4.0

(b)

4 -3.0

Know we need $x \rightarrow 2^-$ in (b) but the latter process wrong

5 -2.0

Wrong on finding $x \rightarrow 2^-$

6 -2.0

Wrong on finding a

7 -1.0

partially wrong on (a)

8 -1.0

Lack of reasoning in (b)
e.g. Should not use substitution 1.99 to find limit

9 -1.0

Minor computational mistake

0 -0.0

number form the other version

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Q8 Question 0

1 point

Rubric !

Math 10A Excel with Integrity Pledge

The Excel with Integrity pledge affirms the UC San Diego commitment to excel with integrity both on and off campus, in academic, professional, and research endeavors.

According to the International Center for Academic Integrity, academic integrity means having the courage to act in ways that are honest, fair, responsible, respectful & trustworthy even when it is difficult. Creating work with integrity is important because otherwise we are misrepresenting our knowledge and abilities and the University is falsely certifying our accomplishments. And when this happens, the UCSD degree loses its value and we've all wasted our time and talents!

Name: _____ PID: _____

Excel with Integrity Pledge

*I am **fair** to my classmates and instructors by not using any unauthorized aids.*
*I **respect** myself and my university by upholding educational and evaluative goals.*
*I am **honest** in my representation of myself and of my work.*
*I accept **responsibility** for ensuring my actions are in accord with academic integrity.*
*I show that I am **trustworthy** even when no one is watching.*

Affirm your adherence to this pledge by writing the following statement in the space below:

I Excel with Integrity.

Signature: _____ Date: _____

1 +1.0

Correct

+ Add Rubric Item

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